HEMA MADHURI MAMIDI

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OBJECTIVE

B.Tech AI Data Science student (CGPA 9.63, Branch Topper) with experience building and deploying full-stack and ML-integrated applications using Python, Django, and React. Strong in DSA and problem-solving, with multiple end-to-end deployed projects.

EDUCATION

B.Tech in Artificial Intelligence and Data Science — *2024 – 2028*
Vignan's Institute of Information Technology (Vignan's IIT), Andhra Pradesh, India — **CGPA: 9.63 / 10**
Branch Topper (Sem 1); Class Rank 1 (Sem 1–2), Rank 2 (Sem 3)

TECHNICAL SKILLS

Languages	Python, C, Java
DSA & OOP	Arrays, Linked Lists, Trees, Graphs, DP, Recursion, Sorting/Searching; OOP design principles
Web Development	HTML5, CSS3, Bootstrap, Django (REST APIs, Authentication, ORM)
AI / ML	TensorFlow, Keras, Transfer Learning, EfficientNetB0, NumPy, Pandas, Matplotlib
Databases	MySQL, SQLite
Developer Tools	Git, GitHub, VS Code, PyCharm, Anaconda, Google Colab

PROJECTS

Nail Health AI — DTI Project | Team Lead — Live Demo | GitHub
Python, Django, TensorFlow, EfficientNetB0

- Spearheaded end-to-end development of an AI-powered healthcare web app for automated nail disease detection, leading a cross-functional team across UI, ML, backend, and deployment.
- Designed and trained an EfficientNetB0 transfer-learning model classifying five nail conditions (pitting, clubbing, melanoma, blue finger, healthy) achieving ∼**94% validation accuracy**.
- Engineered a `DISEASE_INFO` mapping system delivering structured outputs: confidence scores, clinical causes, and personalised dietary recommendations per detected condition.
- Developed workflows for doctor verification, appointment booking, automated medical report generation, and payment gateway integration.

NiyogaX — Voice-First Job Platform — CSP Project (Ongoing) | Team Lead — GitHub
React, Django, Web Speech API

- Leading system design, sprint planning, and task distribution for a voice-first employment platform empowering low-literacy workers through accessible technology.
- Architecting *Niyo*, an AI voice assistant (Web Speech API) with dynamic intent matching and fallback handling for unrecognized input, enabling job discovery, onboarding, and full navigation by voice in Telugu.
- Designing a dual-role system (Worker/Contractor) with voice-guided profiles, category-based job filtering, SOS emergency alerts, and a multilingual contractor dashboard.
- Implementing role-based access control, voice command routing, and progressive onboarding to maximise accessibility for non-technical users.

Wallet Watch — Expense Tracker (Deployed) — Live Demo | GitHub
HTML, CSS, Bootstrap, Python, Django, Chart.js | Render

- Developed a full-stack expense management application with secure user authentication, category-wise CRUD operations, and multi-user data isolation using Django's ORM and session management.
- Built dynamic analytics dashboards with Chart.js visualising pie charts, monthly spending trends, and top expense categories; deployed on Render with CI-friendly GitHub integration.

YouTube Study Mode — Chrome Extension — GitHub
HTML, JavaScript, Chrome Extensions API

- Built a productivity-focused Chrome extension that eliminates YouTube distractions by hiding Shorts, video recommendations, and autoplay via content scripts and DOM manipulation using Chrome Extension APIs.
- Implemented a popup-based toggle UI with persistent state management via Chrome Storage API, ensuring Study Mode preference is maintained seamlessly across browser sessions and page reloads.

CERTIFICATIONS

- **NPTEL — Data Structures and Algorithms using Python** | Silver Medal, Top 5% — *2025*
- **NPTEL — Problem Solving Through Programming in C** — *2025*
- **NPTEL — Cloud Computing** — *2025*

ACTIVITIES & COMPETITIONS

- **Google Solution Challenge 2026** – Global competition for UN SDG-aligned solutions.
- **TechNova Hackathon** (12 hr, ANITS College) — Built frontend for a disaster-response app concept with AI-based image verification to flag fake reports.
- **SuShacks Hackathon** (24 hr, Vignan's IIT) — Built frontend for a hospitality emergency alert system routing incidents to nearby staff instead of reception desk via a live dashboard.
- Practising DSA on LeetCode & HackerRank; exploring predictive modelling and data visualisation with NumPy, Pandas, Matplotlib.